Quirk Table I reproduction

arXiv:1708.02243 · colored m_Q = 1.8 TeV, Λ = 2 keV · EW m_Q = 800 GeV, Λ = 4 keV · results as of 6 October 2026 · Wilson 95% intervals · p = exact binomial against the paper's value

Trigger and fiducial efficiency (the same in every SCT setting)

Benchmark	Sample	k / n	ε [Wilson 95%]	Paper	p	Verdict
ε_{trig} = N(pT of the quirk pair > 200 GeV) / N(events after jet matching) Same cut as the paper (Sec. III B), on the final-state quirk pair.						
Colored	COL1000	101 / 419	0.241 [0.203, 0.284]	0.24	0.95	**No statistically significant inconsistency**
	COL2000	207 / 873	0.237 [0.210, 0.266]	0.24	0.87	**No statistically significant inconsistency**
EW	EW1000	89 / 511	0.174 [0.144, 0.209]	0.10	2.8×10^{-7}	**Strongly inconsistent**
	EW2000	186 / 1045	0.178 [0.156, 0.202]	0.10	1.8×10^{-14}	**Strongly inconsistent**
ε_{fid} = N(both quirks cross all 8 barrel layers inside the [27] lengths, within 25 ns) / N(triggered) The paper's definition (Sec. IV); colored: both R-hadrons charged, in numerator and denominator.						
Colored	COL1000	6 / 27	0.222 [0.106, 0.408]	0.28	0.67	**No statistically significant inconsistency**
	COL2000	22 / 68	0.324 [0.224, 0.442]	0.28	0.42	**No statistically significant inconsistency**
EW	EW1000	33 / 89	0.371 [0.278, 0.475]	0.28	0.059	**Borderline / mild tension**
	EW2000	62 / 186	0.333 [0.270, 0.404]	0.28	0.12	**No statistically significant inconsistency**

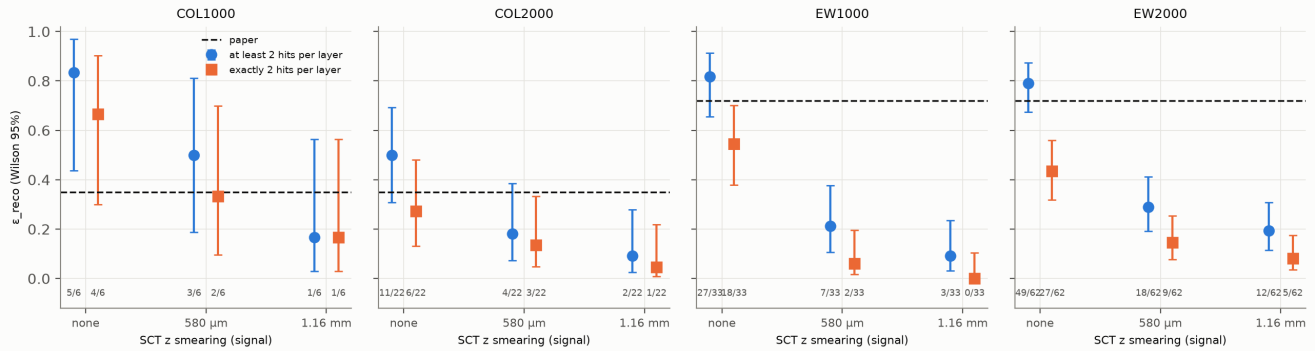
Four ways to set the SCT resolution along the beam (z)

	SCT z: signal / background	ε_{reco} COL2000 (paper 0.35)	ε_{reco} EW2000 (paper 0.72)	Conversion + pile-up in signal region (paper ≈ 2 of 9139)	Total ε COL / EW (paper 0.024 / 0.020)	Signal / background like the paper?
A	no smearing / 580 µm	11/22 = 0.50 [0.31, 0.69] p = 0.18	49/62 = 0.79 [0.67, 0.87] p = 0.26	0 / 9139 (UL 3.3×10^{-4}) cited material model	0.038 / 0.047	**signal ✓ background ✓**
B	580 µm / 580 µm	4/22 = 0.18 [0.07, 0.39] p = 0.12	18/62 = 0.29 [0.19, 0.41] p = 2.8×10^{-12}	0 / 9139 (UL 3.3×10^{-4}) cited material model	0.014 / 0.017	**signal ✗ background ✓**
C	1.16 mm / 1.16 mm	2/22 = 0.09 [0.03, 0.28] p = 0.012	12/62 = 0.19 [0.11, 0.31] p = 1.1×10^{-17}	0 / 9139 (UL 3.3×10^{-4}) no material model	0.007 / 0.011	**signal ✗ background ✓**
D	no smearing / no smearing	11/22 = 0.50 [0.31, 0.69] p = 0.18	49/62 = 0.79 [0.67, 0.87] p = 0.26	146 / 9139 = 1.6% cited material model	0.038 / 0.047	**signal ✓ background ✗**

ε_{reco} here: every layer crossing within 25 ns gives a hit, ≥ 2 hits in 7 layers. Background: converted photon + μ = 50 pile-up, 9139 photons, Ref. [27] lengths, fit from the vertex (unchanged: GeV electrons cross each layer once). Pile-up only (5000 events, 1.16 mm): 0 in the signal region. Total ε uses the 2000-event samples. ✓ = paper value inside our Wilson 95% interval (signal: p ≥ 0.05 for both samples).

Reconstruction efficiency: every SCT setting, both readings of the hit cut

Sample (paper)	SCT z	≥ 2 hits k/n	ε [Wilson 95%]	p	Verdict	exactly 2 k/n	ε [Wilson 95%]	p
ε_{reco} = N(fiducial, plane found, 2 hits in 7 layers + ≥ 1 in the 8th, Δw < 1 cm, Δs < 0.01 cm) / N(fiducial); every layer crossing within 25 ns gives a hit								
COL1000 (0.35)	none	5/6	0.83 [0.44, 0.97]	0.022	**Statistically inconsistent**	4/6	0.67 [0.30, 0.90]	0.19
	580 µm	3/6	0.50 [0.19, 0.81]	0.43	**No significant inconsistency**	2/6	0.33 [0.10, 0.70]	1.00
	1.16 mm	1/6	0.17 [0.03, 0.56]	0.67	**No significant inconsistency**	1/6	0.17 [0.03, 0.56]	0.67
COL2000 (0.35)	none	11/22	0.50 [0.31, 0.69]	0.18	**No significant inconsistency**	6/22	0.27 [0.13, 0.48]	0.51
	580 µm	4/22	0.18 [0.07, 0.39]	0.12	**No significant inconsistency**	3/22	0.14 [0.05, 0.33]	0.042
	1.16 mm	2/22	0.09 [0.03, 0.28]	0.012	**Statistically inconsistent**	1/22	0.05 [0.01, 0.22]	0.0013
EW1000 (0.72)	none	27/33	0.82 [0.66, 0.91]	0.25	**No significant inconsistency**	18/33	0.55 [0.38, 0.70]	0.032
	580 µm	7/33	0.21 [0.11, 0.38]	2.0×10^{-9}	**Strongly inconsistent**	2/33	0.06 [0.02, 0.20]	2.0×10^{-15}
	1.16 mm	3/33	0.09 [0.03, 0.24]	5.5×10^{-14}	**Strongly inconsistent**	0/33	0.00 [0.00, 0.10]	5.7×10^{-19}
EW2000 (0.72)	none	49/62	0.79 [0.67, 0.87]	0.26	**No significant inconsistency**	27/62	0.44 [0.32, 0.56]	2.9×10^{-6}
	580 µm	18/62	0.29 [0.19, 0.41]	2.8×10^{-12}	**Strongly inconsistent**	9/62	0.15 [0.08, 0.25]	5.6×10^{-21}
	1.16 mm	12/62	0.19 [0.11, 0.31]	1.1×10^{-17}	**Strongly inconsistent**	5/62	0.08 [0.03, 0.18]	4.0×10^{-26}



Same numbers as the table. Circles: ≥ 2 hits in 7 layers; squares: exactly 2. Bars = Wilson 95%; dashed = paper.

What we found in the review (and corrected)

- **A quirk often crosses the same layer more than once within 25 ns** (it swings back and forth while moving out slowly). Fiducial events with a repeated crossing: COL2000 13/22, EW2000 31/62. Our earlier code kept only the first crossing. The paper computes “the intersections of the trajectories”, so every crossing is now a hit.
- **The hit cut can be read two ways.** The paper says “All but one layer must contain 2 hits”. With extra crossings a layer can hold 3–5 hits. Reading it as ≥ 2 keeps those events; reading it as exactly 2 removes them (EW drops from about 0.8 to 0.44–0.55). With first crossings only, both readings gave nearly the same numbers.
- **Fig. 2 and 3 redone** in the exact centre-of-mass frame (before: equal lab time and a plane normal from $x_1 \times x_2$, which gave gaps). Fig. 4 now shows the median-Δs event, not a hand-picked one.

What the paper and its references say

Paper, Sec. III A (signal detector)

"The intersections of the trajectories with a simplified model of the ATLAS inner detector are calculated, and the center of each pixel hit is used as the input for our analysis. We hereby use the parameters of the various detector elements as specified in [27]. Specifically, for the pixel detector we use the pixel size rather than the resolution and for the silicon microstrip tracker (SCT) we conservatively assume a resolution of twice the intrinsic accuracy quoted in [27]."

Ref. [27] (ATL-PHYS-PUB-2009-002, in arXiv:0901.0512, p. 18 = PDF p. 42)

"The intrinsic accuracies per module in the barrel are 17 μm (R−φ) and 580 μm (z)"

Paper, Sec. III B (trigger)

"the quirk/anti-quirk pair is essentially always central and sufficiently boosted, such that each quirk will most likely intersect each layer only once."

Paper, Sec. IV (signal region)

"2. All but one layer must contain 2 hits, the remaining layer must contain at least 1 hit 3. $\Delta w < 1.0$ cm and $\Delta s <$ 0.01 cm"

Paper, Sec. III C (Δs and Δw)

"The smallest eigenvalue, Δs^2, then gives the minimum value of d^2 ... Then second eigenvalue, denoted by Δw, then provide a measure of the width of the quirks' oscillations. ... – Second eigenvalue of (11), or the "width", $(\Delta w)^2$"

Paper, Sec. III D (backgrounds)

"We model pile-up by randomly selecting minimum bias events ... processed by the simplified detector described in [34]. ... The e^+e^- pair is then passed through the same detector simulation as described for the pile-up background."

Ref. [34] (arXiv:1612.00850), App. A

"We take the resolution and geometry of each component from the ATLAS technical design report"

Paper, Sec. IV (results)

"the reconstruction efficiency for these cuts can be as high as ∼73%. ... Even though Fig. 5 suggests a few background events after the final selection cut of $\Delta s < 0.01$ cm, we suspect that they can easily be removed"

How each choice rests on these lines

- **$\Delta s = \sqrt{\lambda_1}$ and $\Delta w = \sqrt{\lambda_2}$** of Eq. 11 (the summary list gives $(\Delta w)^2$ as the second eigenvalue). We use this.
- **Every crossing is a hit:** the paper calculates "the intersections"; its "only once" is an expectation, not a cut. Whether a layer may then hold more than 2 hits is not stated.
- **SCT z, scenarios B and C** follow the signal sentence literally: "twice the intrinsic accuracy" of [27] gives 1.16 mm (C); undoubled [27] is 580 μm (B). The paper does not say which direction is doubled.
- **Scenario A** follows the paper's results: no smearing reproduces the ≈ 73% and the Fig. 5 signal shape; the background uses the cited detector [34] (≈ 580 μm from the TDR). No sentence states "no smearing".
- **Scenario D** (no smearing for both) reproduces the signal but gives 146 / 9139 conversions in the signal region, against the paper's "a few".

Definitions

ε_{trig} = N(pT of the quirk pair > 200 GeV) / N(events after jet matching)

ε_{fid} = N(triggered, both quirks cross all 8 barrel layers inside the [27] lengths, within 25 ns) / N(triggered)

ε_{reco} = N(fiducial, plane found, 2 hits in 7 layers + ≥ 1 in the 8th, $\Delta w < 1$ cm, $\Delta s < 0.01$ cm) / N(fiducial); every layer crossing within 25 ns gives a hit

$\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$ (paper Eq. 12)

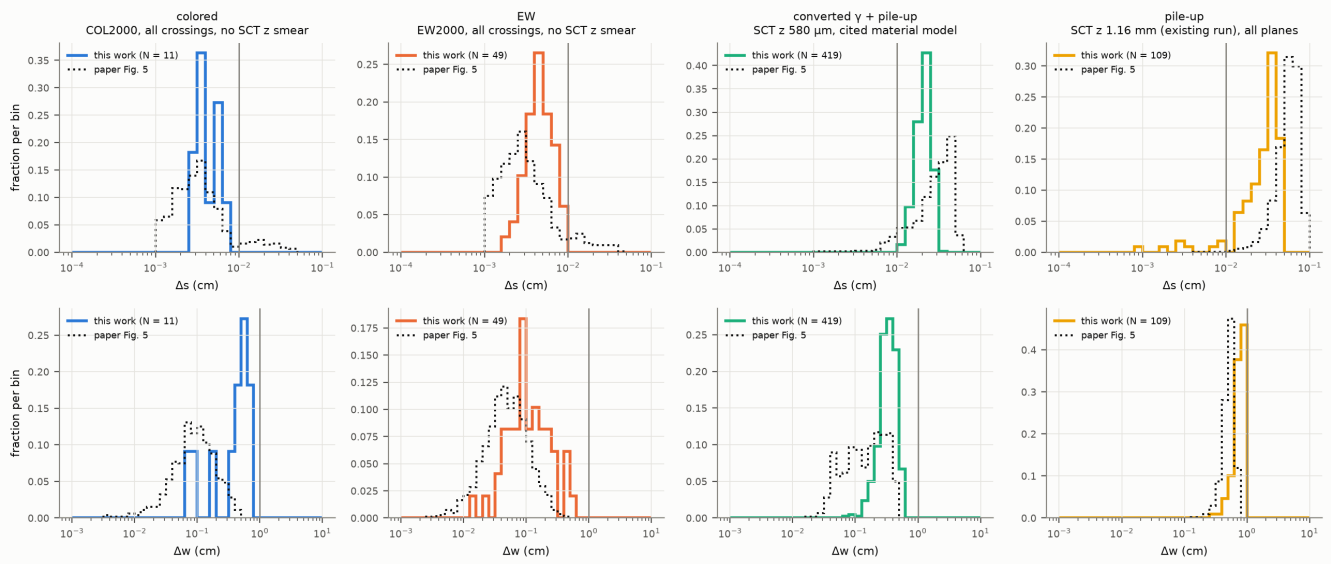
Colored: both R-hadrons must be charged; applied to the triggered events before ε_{fid} (27 of 101 and 68 of 207 events).

Wilson 95% interval for k of n ($\varepsilon = k/n$, $z = 1.96$): $[\ \varepsilon + z^2/2n \pm z\sqrt{(\varepsilon(1-\varepsilon)/n + z^2/4n^2}\ ]\ /\ (1 + z^2/n)$

95% upper limit (UL) when 0 of n pass: $UL = 1 - 0.05^{1/n} \approx 3/n$

p = exact binomial probability, if the paper's value were exactly right, of a count as unlikely as ours or more

Paper Figure 5: Δs and Δw before the Δs cut



Unit area on the paper's bins; dotted = paper's Fig. 5. Signal: plane + hit cut (≥ 2), every crossing, no SCT z smearing, before the Δs cut (Δw < 1 is part of the paper's plane search, so no event above 1 cm can appear). Converted γ + pile-up: SCT z 580 μm, cited material model. Pile-up: existing run (1.16 mm), all planes.

- **Signal Δs:** median 38 μm (colored) and 45 μm (EW), against about 35 and 28 μm in the paper. All our signal events after the plane and hit cut already have Δs < 0.01 cm; the paper has a tail above it (8–10%).
- **Signal Δw:** median 0.49 cm (colored) and 0.10 cm (EW), against about 0.09 and 0.045 cm in the paper. EW is about 2× wider, colored about 5×. Using ½√λ₂ would explain the EW factor but not the colored one.

Paper Figures 2 and 3

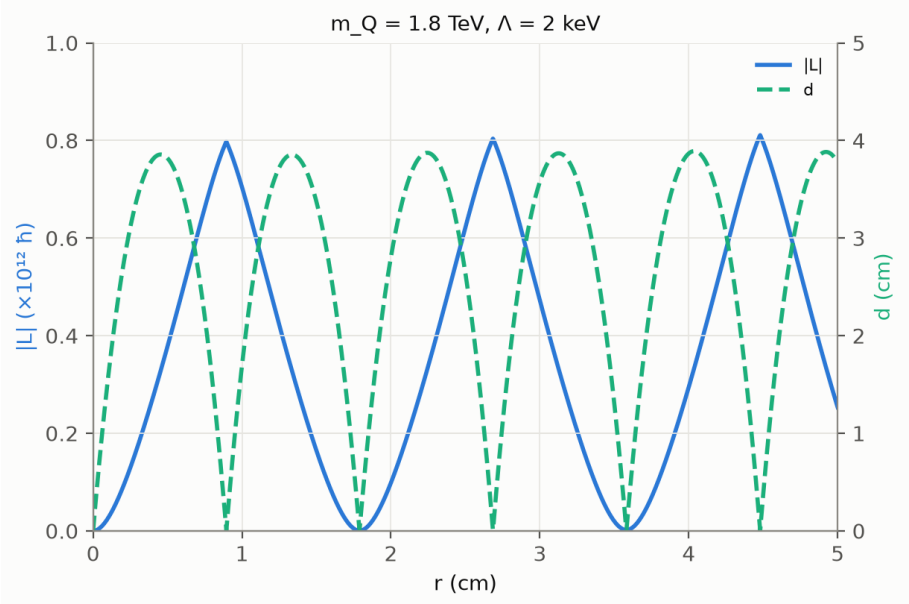


Fig. 2 on the paper's axes: |L| (magnitude of the angular momentum, solid) and separation d (dashed) in the exact centre-of-mass frame, vs the radius r of the centre of mass. Colored 1.8 TeV, 2 keV, COL2000 fiducial event with the median $|L|_{max}$. Where |L| ≈ 0, d = 0.017–0.019 cm (≫ $1/m_Q$), so the pair does not annihilate. |L| is zero at production and grows like 1 − cos(ωt), so it peaks at every second minimum of d; the paper's single event shows a different pattern.

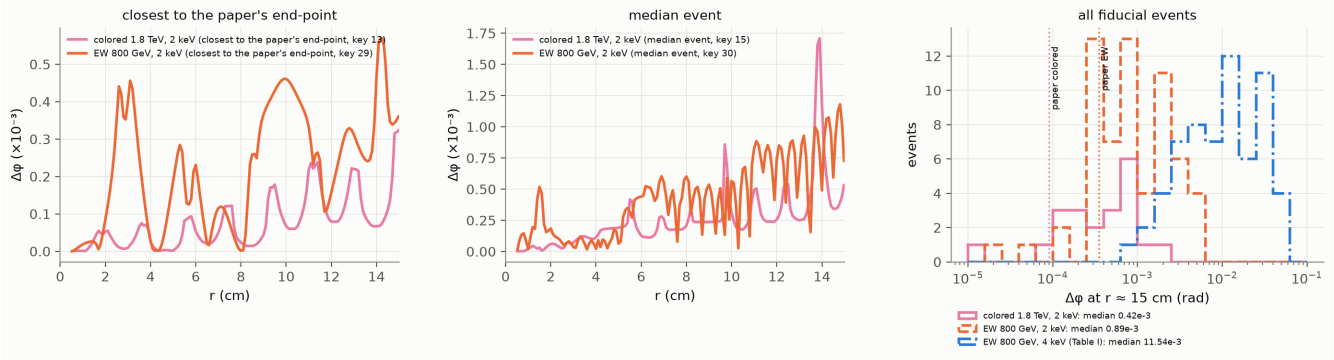
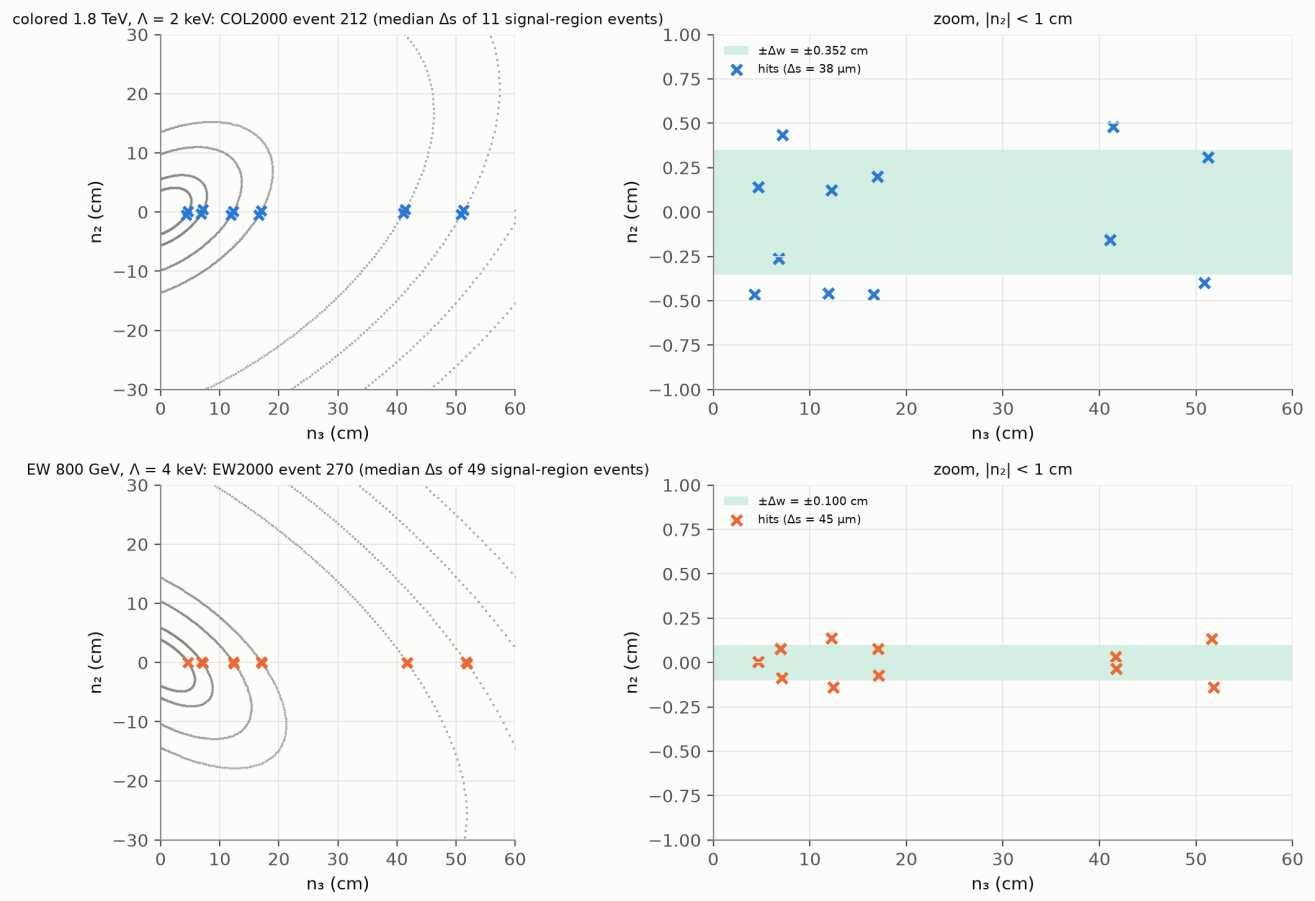


Fig. 3 on the paper's axes (r = 0–15 cm, Δφ in 10^{-3} rad): rotation of the plane of relative motion (normal from $\vec{L}$; a sliding plane fit gives the same within 10%). Left: event closest to the paper's curve; middle: median event; right: Δφ at 15 cm for all fiducial events. Paper values (dotted): colored ≈ 0.09×10^{-3}, EW ≈ 0.35×10^{-3}. Our medians: colored 0.42×10^{-3}, EW (2 keV) 0.89×10^{-3}; the paper's values sit at our 9th and 23rd percentile. EW 4 keV (Table I benchmark): 11.5×10^{-3}.

Paper Figure 4: median event, not hand-picked



Selection rule: among events that pass the full signal region (every crossing, no SCT z smearing, ≥ 2 hits), the one with the median Δs. Colored: COL2000 event 212 (6th of 11, Δs = 38 μm, Δw = 0.35 cm). EW: EW2000 event 270 (25th of 49, Δs = 45 μm, Δw = 0.10 cm). Left: all hits in the plane's (n_3, n_2) coordinates with the layers as ellipses; right: zoom with the ±Δw band. Paper's example: Δw ≈ 0.074 cm.